

Questions

Questions from Graduation Project Plan 2024-2025

Section 1: The Assignment

Assessor notes by Christiaan Verhoef (teacher, Value Chain Hackers, Windesheim)

VCH faces a challenge common to many research organizations: valuable project results often disappear when students move on, and promising ideas rarely become tools that companies can use. This undermines trust among partners and limits the real-world impact of VCH's work. To address this, VCH aims to design a reproducible research process to strengthen its findings and to increase its potential impact.

Why Reproducibility Matters

Reproducibility is the foundation of reliable science. As highlighted by Wellcome Open Research, reproducible research strengthens scientific evidence and the reliability of results. When others can reproduce a study, it lends credibility to the findings and increases the chance that those results will inform policy or business practice. Conversely, irreproducible research erodes trust, both within the scientific community and among external partners. This trust is vital, especially when public funding and real-world decisions depend on research outcomes¹.

Moreover, reproducibility brings efficiency. When research is well-documented and transparent, future teams can reuse data, code, and methods—saving time and preventing the loss of knowledge when original contributors leave. It also helps minimize misinformation, as transparent methods make it easier for reviewers and collaborators to spot mistakes before they become entrenched in the literature¹.

Companies that fail to embrace these practices risk losing credibility with partners, missing out on valuable collaborations, and facing increasing regulatory scrutiny. Conversely, early adopters of reproducible workflows gain competitive advantages through enhanced transparency, reduced risk, and accelerated innovation cycles. By starting with pilot projects, demonstrating clear business value, and providing support for change, organizations can overcome resistance and position themselves for long-term success in an evolving research ecosystem that increasingly demands openness and accountability.

Tools and Systems that Enable Reproducibility

Modern workflow management systems and collaborative platforms are key to making reproducibility practical across teams and projects. For example, GitHub has proven to be an effective platform for organizing, documenting, and sharing every stage of a research project—from experiment design to data analysis and manuscript writing. By adopting software development workflows, research teams can ensure that all aspects of their work are version-controlled, transparent, and accessible to future collaborators. This approach not only increases the efficiency of knowledge transfer but also supports large, cross-institutional collaborations².

Workflow Management Systems (WMS) further support reproducibility by standardizing how experiments and analyses are conducted and documented. A recent study demonstrated that modular WMSs, designed around principles like Findable, Accessible, Interoperable, and Re-usable (FAIR), make it easier for researchers to share protocols, models, and data. This fosters open collaboration and addresses the reproducibility crisis by ensuring that knowledge and tools remain accessible and usable long after the original team has moved on³.

Building Trust and Accountability

Trust in research is not just about technical reproducibility—it's about transparency and accountability. The EU-funded TRUSTparency project emphasizes that trust in science depends on transparent, collaborative processes that allow knowledge to self-correct. By co-creating reproducibility promotion plans and involving all stakeholders (including companies, regulators, and students), organizations can develop concrete steps to embed reproducibility into their everyday work. This collective approach empowers institutions to monitor and improve their practices, building a culture where research is both trusted and impactful⁴.

Frameworks for Assessing and Improving Reproducibility

Tools like the RepeAT framework provide concrete ways to assess and improve research transparency and accessibility. By operationalizing key concepts—such as research design, data sharing, and documentation—RepeAT enables organizations to benchmark their practices and make targeted improvements. Widespread adoption of such frameworks helps ensure that research outputs remain available and comparable across domains and institutions⁵.

Assignment topic: VCH proposes to innovate the research process used within Value Chain Hackers (VCH), with the aim of enabling reproducible research workflows. The project will apply the Enterprise Data Reference Model (EDRM) to design and implement this reproducible process. The goal is to create a scalable and repeatable research methodology that can be used across future VCH projects and academic-industry collaborations.

This assignment has multiple dimensions:

- Convincing the other lab-managers of the high need of reproduceable research.
- Making it easy enough for the assistant researchers (students) to use these tools and improve their research, instead of slowing it down.
- Showing companies that this method leads to real validation for the projects.

The main measurement of this assignment is:

How many projects exist and still continue after the original researchers have left?

In addition, the project will validate the need for this innovation and improve the chances that VCH's projects:

Make real impact in the world, not just remain as good ideas.

Breach institutional barriers and influence corporate practices.

Create mechanisms that allow accountability when companies break laws or undermine sustainability efforts.

1. **Does the assignment justifiably constitute an innovation assignment large enough to justify 30 EC worth of individual work (including preparation and research)?**

This will be accomplished by carrying out a complete innovation cycle: the student will conduct a literature review and stakeholder interviews (define), design an EDRM-aligned reproducibility process (design), implement it in a real lab context (execute), and validate it through feedback and impact assessment (learn). The project will require significant original work and leadership by the student across multiple fields (process innovation, data management, organizational change), over 8 months, which more than justifies the 30EC worth of effort.

2. **Does the assignment explicitly show that it is a clear learning opportunity compared to what the student has done before?**

The learning will be accomplished by requiring the student to master and apply new concepts and frameworks: EDRM, formal reproducibility standards, stakeholder alignment in institutional contexts, and legal aspects of transparency and compliance. The student will also develop leadership skills by managing the full project cycle in a complex stakeholder environment learning not yet demonstrated in previous assignments.

3. **Is the assignment clear and concise in terms of what needs to be achieved?**

The project will achieve its goals through the following steps:

- Analyse existing gaps in VCH reproducibility (define).
- Design a reproducible research process and supporting toolchain based on EDRM (design).
- Implement and pilot the process in live VCH projects (execute).
- Evaluate adoption and iterate improvements (learn).
- Deliver final documentation, training materials, and adoption recommendations to the client.

4. **Does the assignment fit the student's graduation track?**

The fit will be demonstrated by applying core Business Innovation skills: designing and embedding an innovative organisational process, facilitating cultural change, and creating impact in a real-world setting that crosses business, academic and legal domains. The assignment fully reflects the innovation consultant role expected of this track.

5. **Does the commissioning client meet the requirements set in the graduation project handbook?**

This requirement is fulfilled as follows: Value Chain Hackers is a recognised applied research lab under the Windesheim Lectoraat Supply Chain Finance, with a professional structure, funded projects, and partnerships — formally meeting the client criteria for this graduation track.

6. ****Does the execution of the assignment include:**

- Extensive research (define)
- Identifying and developing solutions (design)
- (Partially) implementing solutions (execute)
- Monitoring/evaluating (learn)
- Dealing with others (team and/or stakeholders)
- Self-management (lead)?**

Each part will be accomplished as follows:

- **Define:** The student will conduct a literature review on reproducibility and EDRM, plus stakeholder interviews to map current gaps and needs.
- **Design:** A tailored reproducibility process will be designed and documented, with tools selected and workflows mapped.
- **Execute:** The new process will be implemented in two active VCH projects, including hands-on setup of tools and processes with teams.
- **Learn:** The student will monitor usage, collect structured feedback, and analyse adoption and outcomes to refine the process.
- **Deal with others:** The student will manage collaboration with VCH staff, students, academic partners, and industry stakeholders to ensure alignment and buy-in.
- **Lead:** The student will manage the full project plan, communications, time planning and delivery, demonstrating independent leadership throughout.

7. **Does the student provide a clear, relevant and credible (substantiated) analysis of the past, current, and near future situation at micro, meso and macro level, leading to the necessity/desirability of the assignment?**

This analysis will be accomplished by:

- Reviewing VCH's historical project records (micro) to understand loss of research value due to lack of reproducibility.
- Interviewing staff and partners to document current process gaps and barriers (meso).
- Studying policy trends (macro) around transparency, accountability and EU legal drivers, to connect VCH's future needs with external expectations.

These inputs will form the evidence base for the project's design decisions.

8. **Has the student built a credible argument for why this assignment is the right one to address the issue?**

The argument will be demonstrated through a combination of stakeholder evidence (interviews and feedback), benchmarking against best practices in reproducibility and data governance, and impact testing in the VCH lab. By showing that the EDRM-based approach addresses both

VCH's internal problems and external market requirements, the project will justify why this is the right innovation to pursue at this time.
Without this project, VCH's risk of research loss, lack of credibility, and low industry adoption will persist thus limiting its long-term impact."

Section 2: The Research Design

Assessor notes by Christiaan Verhoef (teacher, Value Chain Hackers, Windesheim)

Assignment topic: VCH proposes to innovate the research process used within Value Chain Hackers (VCH), with the aim of enabling reproducible research workflows. The project will apply the Enterprise Data Reference Model (EDRM) to design and implement this reproducible process.

1. **Is the planned research multifaceted and does it justify an individual workload of 6 EC (including the research design and the research report)?**

The student will accomplish this by conducting a multifaceted study that includes:

- A desk study of reproducibility frameworks, EDRM literature, and EU regulatory requirements.
 - Field research through interviews with internal and external stakeholders (students, staff, partners).
 - Process analysis of current VCH workflows.
 - Pilot implementation of a new reproducibility process, followed by evaluation.
- These combined tasks will produce sufficient depth, complexity, and workload to justify 6 EC.

2. **Are the main research question and sub questions appropriate considering the assignment?**

The student will accomplish this by formulating the following main question:

How can the Enterprise Data Reference Model be applied to design and implement a reproducible research process for VCH?

Sub-questions will include:

- What gaps currently exist in reproducibility within VCH?
- Which elements of EDRM best apply to the VCH context?
- What are the barriers and enablers to process adoption?
- What technical and organisational changes are required to achieve sustainable reproducibility?

3. **Are the questions mainly obvious, with some insightful ones, demonstrating ability to question in different kinds of ways?**

This will be achieved by balancing sub-questions: some address obvious gaps (tool selection, process gaps), while others go deeper into human factors (adoption barriers) and strategic questions (alignment with EU trends), demonstrating multi-level questioning.

4. **Does the student develop relevant research methods (including research population) for each of the sub questions, addressing all 5W1H for each question?**

The student will accomplish this by mapping each sub-question to specific methods:

- Desk research for what/why questions.
- Semi-structured interviews for who/why/how questions (targeting VCH team, students, external partners).
- Observational studies of current processes (how/when/where).
- Pilot testing for how questions (process implementation and feedback).

The plan covers all 5W1H dimensions.

5. **Is the methodology justified based on the data sought?**

The student will demonstrate this by selecting qualitative methods (interviews, observation) to capture current human and organisational factors, and combining this with desk research to align with best practices and legal frameworks — a justified combination for this type of innovation project.

6. **Does the student formulate relevant hypotheses regarding the research questions and appropriate ways to test them?**

The student will define hypotheses such as:

- Applying an EDRM-based process will increase the reproducibility and reusability of VCH research outputs.
- Tool-supported workflows will improve adoption rates among new VCH researchers.

These will be tested through pre- and post-intervention evaluation of research outputs and user feedback.

7. **Does the student develop an appropriate plan/method to process and analyse the data collected?**

The student will process qualitative interview data using thematic analysis. Pilot project outcomes will be evaluated using structured assessment criteria (completeness of documentation, ease of reuse). Process observations will be synthesised into design recommendations.

8. **Is the planned research realistically feasible in terms of target group, time, and resources?**

The student will accomplish this by targeting current VCH staff and project teams already active in the lab this semester, ensuring access to the target group. Time planning follows an iterative sprint model that fits the semester schedule, with committed support from VCH leadership.

9. **Has the student set up an effective system to file (and share) relevant information found along the way for later use?**

The student will implement this through a version-controlled knowledge base using GitHub and Quarto, ensuring that all research data, process documentation, and design iterations are transparently recorded and reusable by future VCH members.

Section 3: The Project Design

Assessor notes by Christiaan Verhoef (teacher, Value Chain Hackers, Windesheim)

Assignment topic: VCH proposes to innovate the research process used within Value Chain Hackers (VCH), with the aim of enabling reproducible research workflows. The project will apply the Enterprise Data Reference Model (EDRM) to design and implement this reproducible process.

1. **Does the project plan provide a clear and substantiated scope of the project?**

The student will accomplish this by defining a clear project scope:

- Design and pilot a reproducible research process based on EDRM.
- Implement and test this process in live VCH projects.
- Deliver tools, documentation, and training for future use.

This scope is documented in the project plan and will be validated through each project sprint.

2. **Does it identify relevant criteria and specs for the end results of the project, based on the client brief and student’s own investigation of the situation?**

The student will establish criteria such as:

- Research outputs should meet reproducibility standards (documented workflows, traceable data).
- The process must be adoptable by students and external partners.
- Outputs must align with EU transparency requirements.

These specs were derived from discussions with VCH staff and analysis of best practices.

3. **Does it justify the feasibility of these specs and explain when desired specs will not be met?**

The student will address feasibility by mapping critical constraints (e.g. student skill level, tool complexity) and adjusting the process accordingly.

Any specs that cannot yet be fully met (such as automated metadata generation) will be flagged and justified in the final report.

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4. **Are the scope, criteria and specs defined in a way that they can serve as an anchor throughout the project, for both the student and the client?**

This will be achieved through a published criteria checklist that will be referenced during all project reviews and sprint evaluations, ensuring alignment between student and client expectations.

5. **Have the primary stakeholders been identified? Are they relevant and complete?**

Stakeholders will be mapped as:

- Internal: VCH staff, students in current projects.
- External: Academic partners, corporate collaborators.

This stakeholder map will be reviewed with the client during the first sprint.

6. **Is it correctly identified what is at stake for each stakeholder and what active role they will play in the project (if any)?**

The student will document for each stakeholder group:

- What value reproducibility brings to them.
- What input or feedback they are expected to provide.

This will be tracked in a stakeholder engagement plan.

7. **Has the availability of these stakeholders been assessed when their involvement is needed?**

The student will schedule interviews and feedback sessions in coordination with active VCH project cycles, ensuring availability is aligned with project needs.

8. **Is the list of necessary resources to execute the project complete and relevant?**

The project requires:

- Access to VCH project data.
- EDRM framework materials.
- Tools (GitHub, Quarto, JupyterHub).
- Stakeholder time for interviews and testing.

All are committed by the lab or readily available.

9. **Has the feasibility of acquiring these resources when needed been assessed?**

Resource needs were validated with the lab coordinator and IT support prior to project approval; availability is secured.

10. **Have all the necessary deliverables been identified, including their format, timing, and audience?**

Deliverables include:

- Process design (documented in Quarto).
- Supporting toolchain (GitHub repository).
- Pilot evaluation report.
- Final reproducibility guide and training material.

Each is linked to the semester sprint schedule and reviewed with VCH staff.

11. **Have all necessary tasks to execute the project been identified, including who is responsible for what?**

The student will use a detailed task board (GitHub Projects) identifying all project tasks, with clear ownership and timeline responsibilities assigned.

12. **Are the tasks, deliverables and resources planned and prioritised appropriately to create a feasible planning for the entire graduation project period?**

The project follows a sprint-based structure with progressive prioritisation:

- Sprint 1: Define and scope.
- Sprint 2–3: Design and pilot.
- Sprint 4: Evaluate and refine.
- Sprint 5: Finalise and deliver.

This ensures pacing is realistic and achievable within the semester.

13. **Is there an appropriate scaffolding of these tasks, considering their interrelationships and dependencies?**

Dependencies (e.g. pilot feedback required before final documentation) are mapped in the project board and reflected in sprint planning.

14. **Have appropriate continuous feedback loops with relevant parties been incorporated?**

Feedback loops will include:

- Bi-weekly check-ins with the VCH core team.
- End-of-sprint reviews.
- User testing feedback from students and partners.

This ensures iterative development and learning.

15. **Are there relevant criteria and specs for the progress of the project, and is there an appropriate system for the student, supervisor, and client to follow progress?**

Progress will be tracked on the GitHub Project board, with clear criteria for sprint completion and milestone achievement. This system is visible to the student, supervisor, and client.

16. **Have obvious risks in the execution of the project been identified and are appropriate ways to deal with them suggested?**

Identified risks include:

- Stakeholder resistance to process change.
- Complexity of toolchain integration.

Mitigations include:

- Early involvement of stakeholders in process design.
- Selection of user-friendly, well-supported tools.
- Incremental implementation to build acceptance.